

AUTOSAR High-Level Design (HLD) Document

System Architecture: Body Control Module (BCM) Subsystem | Release:
Rel_4.4.0_BCM_v1.2

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AUTOSAR Release	Classic Platform 4.4.0	Safety Level	ISO 26262 ASIL-B
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1. Executive Architectural Overview & Scope

This High-Level Design document specifies the AUTOSAR software architecture for the Body Control Module (BCM). The BCM ECU oversees comfort, lighting, vehicle access control, wiper logic, and power distribution across the vehicle chassis. The software architecture strictly adheres to AUTOSAR Classic Release 4.4.0 layered structure, separating the Application Software layer (ASW), Runtime Environment (RTE), and Basic Software (BSW). The physical microcontroller target is the Renesas RH850/F1KM-S4.

Primary architectural objectives: (a) modular software components with well-defined sender-receiver and client-server interfaces, (b) strict timing determinism with cyclic 10ms, 20ms, and 100ms task scheduling, (c) complete diagnostic event reporting through DEM, and (d) seamless CAN FD and LIN network communication abstraction via Com and PduR modules.

2. Software Component (SWC) Inventory

The Application Software Layer comprises four principal Application Software Components (Atomic SWCs) and one Complex Device Driver (CDD):

SWC Name	SWC Type	Periodicity	ASIL Level	Description & Responsibility
SWC_ExteriorLighting	ApplicationSWC	10 ms	ASIL-B	Controls high beam, low beam, turn indicators, DRL, and emergency hazards.
SWC_CentralLocking	ApplicationSWC	20 ms	ASIL-A	Manages smart keyless entry, door latches, child safety locks, and anti-theft immobilizer handshake.
SWC_WiperControl	ApplicationSWC	20 ms	QM	Interprets rain sensor feedback, driver stalk position, and controls front/rear wiper speeds.
SWC_PowerManagement	ApplicationSWC	100 ms	ASIL-B	Monitors 12V battery voltage, executes ignition power state transitions (OFF/ACC/RUN/CRANK), and initiates load shedding.

CDD_SmartSmartActuator	ComplexDeviceDriver	5 ms / ISR	ASIL-B	Direct hardware driver for smart high-side power switches with integrated current sensing.
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3. Ports and Interface Mapping

Inter-component communication and BSW access occur exclusively via AUTOSAR Ports typed by Sender-Receiver (S/R) or Client-Server (C/S) Interfaces. The table below details all formal port connections:

Source Port	Provider SWC	Interface Name	Type	Target Port	Consumer SWC
P_LightCmd	SWC_ExteriorLighting	If_LightingCommand	S/R	R_LightCmd	CDD_SmartSmartActuator
P_LockStatus	SWC_CentralLocking	If_DoorLockState	S/R	R_LockStatus	SWC_ExteriorLighting
P_PowerMode	SWC_PowerManagement	If_PowerMode	S/R	R_SysPowerMode	SWC_CentralLocking
P_PowerMode	SWC_PowerManagement	If_PowerMode	S/R	R_PowerMode	SWC_ExteriorLighting
P_WiperCmd	SWC_WiperControl	If_WiperCommand	S/R	R_WiperActuator	CDD_SmartSmartActuator
P_DiagEvent	SWC_ExteriorLighting	If_DiagnosticEvent	C/S	R_DiagEvent	BSW_DEM (Diag Event Mgr)
P_DiagEvent	SWC_PowerManagement	If_DiagnosticEvent	C/S	R_DiagEvent	BSW_DEM (Diag Event Mgr)
P_NvMReadWrite	SWC_CentralLocking	If_NvMBlockAccess	C/S	R_NvMBlockAccess	BSW_NvM (Memory Stack)

4. Signal Definitions and Data Dictionary

Signals transmitted across Sender-Receiver interfaces are bound to AUTOSAR Implementation Data Types (IDT). Integrity is validated through range limits and error indicator states.

Signal Name	Interface	Data Type	Valid Range	Init / Default Value	CAN Signal Mapping
Sig_HeadlightState	If_LightingCommand	uint8 (Enum)	0:OFF, 1:LOW, 2:HIGH, 3:AUTO	0 (OFF)	BCM_CAN_Frame_0x120.S1
Sig_TurnIndicatorReq	If_LightingCommand	uint8 (Enum)	0:NONE, 1:LEFT, 2:RIGHT, 3:HAZARD	0 (NONE)	BCM_CAN_Frame_0x120.S2
Sig_VehicleLockStatus	If_DoorLockState	boolean	0:UNLOCKED, 1:LOCKED	1 (LOCKED)	BCM_CAN_Frame_0x240.S1
Sig_PowerSupplyMode	If_PowerMode	uint8 (Enum)	0:SLEEP, 1:STANDBY, 2:ACTIVE, 3:CRANK	0 (SLEEP)	Internal RTE Only
Sig_WiperSpeedCmd	If_WiperCommand	uint8	0:OFF, 1:INT, 2:LOW, 3:HIGH	0 (OFF)	LIN_Frame_Wiper_0x05.S0

5. RTE Runnables, Trigger Events, and Scheduling

Runnables encapsulate the algorithmic execution logic inside SWCs. Runnables are triggered by RTE TimingEvents or DataReceivedEvents.

SWC Name	Runnable Entity	Trigger Event	OS Task Mapping	WCET (Max)
SWC_ExteriorLighting	Runnable_Lighting_Step	TimingEvent_10ms	Task_Periodic_10ms	1.25 ms
SWC_ExteriorLighting	Runnable_Hazard_Event	DataReceivedEvent (HazardReq)	Task_Event_Priority1	0.45 ms
SWC_CentralLocking	Runnable_Locking_Step	TimingEvent_20ms	Task_Periodic_20ms	2.10 ms
SWC_WiperControl	Runnable_Wiper_Step	TimingEvent_20ms	Task_Periodic_20ms	0.90 ms
SWC_PowerManagemen t	Runnable_Power_Step	TimingEvent_100ms	Task_Periodic_100m s	3.40 ms

6. Basic Software (BSW) Stack & Services Integration

The BSW layer consists of standardized service, system, and hardware abstraction modules according to AUTOSAR Classic 4.4:

- **Communication Stack (ComStack):** CAN Interface (CanIf), CAN Transport Protocol (CanTp), PDU Router (PduR), and COM module. Signals received from the Powertrain CAN FD bus (500 kbps / 2 Mbps payload) are unpacked by COM and delivered to the RTE via R-Ports.
- **Diagnostic Stack:** Diagnostic Event Manager (DEM), Diagnostic Communication Manager (DCM), and Function Inhibition Manager (FIM). SWCs report DTC status (e.g., DTC 0xB10014: Headlight bulb short to ground) through Client-Server interface *If_DiagnosticEvent*.
- **Memory Stack (MemStack):** Non-Volatile Memory Manager (NvM), Memory Abstraction Interface (MemIf), and Flash EEPROM Emulation (Fee). Used for storing vehicle configuration parameters, key fob encryption rolling codes, and DTC freeze frame logs.
- **System Services & Mode Management:** BSW State Manager (BswM) and ECU State Manager (EcuM). Coordinates startup phase, RUN mode, and low-power SLEEP mode transition upon ignition off and network silence.

7. Functional Safety (ISO 26262) & ASIL Allocation

Safety goals defined for the BCM subsystem under ISO 26262 include:

1. **SG-BCM-01:** Prevent unintended total loss of exterior lighting during nighttime driving (ASIL-B). Fault detection time interval (FDTI) = 50ms.
2. **SG-BCM-02:** Prevent inadvertent door unlocking at vehicle speeds exceeding 15 km/h (ASIL-B). Redundant vehicle speed verification via CAN message from ESP.
3. **SG-BCM-03:** Mitigate battery thermal runaway or parasitic drain by terminating quiescent current leakage within 300ms during SLEEP mode (QM).

8. Architectural Open Issues and Review Notes

During preliminary architectural review, the following notes and potential inconsistencies were recorded:

- **Issue-BCM-101:** Port *R_AmbientLightSensor* in *SWC_ExteriorLighting* is currently unconnected (dangling port) as the LIN rain/light sensor interface specification is pending supplier confirmation.
- **Issue-BCM-102:** The signal *Sig_TurnIndicatorReq* has a configured data width of 8 bits in the Application layer, whereas CAN Matrix v2.1 defines signal *BCM_CAN_Frame_0x120.S2* as a 4-bit bitfield. Signal scaling/masking must be verified in the COM layer.
- **Issue-BCM-103:** *CDD_SmartSmartActuator* requires direct ISR execution outside the standard 10ms RTE schedule. Memory partition isolation (OS Application non-trusted vs trusted) must be audited.